GENERALIZED ESTIMATION SYSTEM

Version 2.03

Methodology Guide – Influential Data

October 2019

G-Est

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1 Purpose

This document is the Methodology Guide for the Influential Data Module in G-Est, version 2.03. The companion document is the User Guide.

G-Est is the new generalized estimation system of Statistics Canada since 2012.

Influential units are sampling units that have an excessive effect on the estimates. The purpose of the current Module is to find out if there are influential units and, if so, to correct them in some way. It is important to emphasize here that the Module's corrections are done on the weights, and not on the variables of interest.

2 Summary

Influential units are sampling units that have an excessive effect on the estimates. The Module detects these units, if any, and corrects their weights. The weight correction or transformation is done in order to satisfy some criteria: weight conservation at the data group level, removal of influential behavior, bias control and mean squared error reduction.

The weights of influential units will be reduced, but the weights of ordinary units will be increased to compensate. It may happen that the only weight transformation that removes the influential behavior from previously flagged influential units has the consequence of generating influential behavior among ordinary units. If this is so, then these *collateral* influential units are added to the pool of influential units and the Module restarts the process of finding the right weight transformation. The process can – in fact will – be iterative.

In some cases, the user does not want some units to be subject to any weight modification. An example could be take-alls. The Module offers many conditions to identify such excluded units; this identification is done prior to any detection of influential units and is done only once; it is not part of the iterative process. The remaining units, called the modifiable units, are composed of influential units and ordinary units and are the only ones whose weights can be changed.

When the optimal weight transformation is found, the output weights are created. If in addition the user has input bootstrap weights, the Module will create output bootstrap weights by applying the same weight transformation to the input bootstrap weights.

The description above is a summary of the regular execution of the Module. In some cases, the user may already have some initial lists of influential units and excluded units. If so, the Module will accept them. When an initial list of influential units is provided, the user may decide not to activate the detection of influential behavior, using only other criteria to choose the right weight transformation.

Of course, if there are no influential units (either from an initial list or detected by the Module itself), the Module will set the output weights equal to the input weights.

The main algorithm is as follows. Its different steps will be described in more details in the following sections of this document.

1. Initialization of the data group.
2. Identification of excluded units.
3. Detection of influential behavior, using the input weights.
4. Iterative process to find the right weight transformation. At each iteration some units may transit from the ordinary type to the influential type.
5. Creation of the output weights and the output bootstrap weights.

These steps will be described in their order. But before, we will set the context along with its notation, and define exactly what is meant by influential behavior.

3 Context

Let s be a sample for a population U and denote $k \in s$ a unit in the sample.

Consider a partition of data groups U_g, $1 \leq g \leq G$, with $G \geq 1$, of the population U, and let s_g represent the sample part of these data groups:

$$U = \biguplus_{g=1}^{G} U_g , \qquad s_g = U_g \cap s, \qquad s = \biguplus_{g=1}^{G} s_g . \tag{3.1}$$

Let a_k be the initial weight of unit k. Optionally, let $a_{(b)k}$ be initial bootstrap weight of unit k for replicate b, $1 \leq b \leq B$.

Let y_1, ..., y_J, where $J \geq 1$, be analysis variables, or variables of interest, that is, variables whose estimates are of interest.

The population parameters considered are

$$T_{gj} = \sum_{k \in U_g} y_{jk} , \qquad 1 \leq j \leq J, \qquad 1 \leq g \leq G. \tag{3.2}$$

Note that there could be many variables of interest ($J \geq 1$), but there can be only one partition of data groups. For instance the partition could be the various provinces of Canada, or the partition could be the various industrial codes, or the partition could be the cross-values of province and industrial code, but we cannot consider at the same time the partition of provinces *and* the partition of industrial codes (which would be working with marginals).

Using the input weights a_k, we consider the estimates

$$\hat{T}_{agj} = \sum_{k \in s_g} a_k y_{jk}\,, \qquad 1 \le j \le J, \qquad 1 \le g \le G. \tag{3.3}$$

Each unit k in s_g contributes to the value $\hat{T}_{agj}$, $1 \le j \le J$. When the presence of a unit k in s_g has an excessive effect on these estimates, we say that the unit exhibits an influential behavior. What we mean by *excessive effect* will be more rigorously defined in section 4.

The Influential Data Module intends to detect this influential behavior, if any, and remove it by creating a new set of weights $\{w_k : k \in s\}$ in replacement of $\{a_k : k \in s\}$. With these new weights, we get new estimates for the population parameters:

$$\hat{T}_{wgj} = \sum_{k \in s_g} w_k y_{jk}\,, \qquad 1 \le j \le J, \qquad 1 \le g \le G. \tag{3.4}$$

Units deemed influential will get a weight decrease, but the remaining units might see a weight increase to compensate. There could also be some units from s_g that one does not want to be affected by any weight change; we call them excluded units. Therefore we can divide the data group s_g into three parts: $s_{g,\mathrm{INFL}}$ is the list of influential units, $s_{g,\mathrm{EXCL}}$ is the list of excluded units and $s_{g,\mathrm{ORD}}$ is the list of ordinary units, that is, units that are neither influential nor excluded:

$$s_g = s_{g,\mathrm{EXCL}} \uplus s_{g,\mathrm{ORD}} \uplus s_{g,\mathrm{INFL}}\,. \tag{3.5}$$

The general relation between w_k and a_k is

$$w_k \begin{array}{lll} = & a_k, & k \in s_{g,\mathrm{EXCL}} \\ \ge & a_k, & k \in s_{g,\mathrm{ORD}} \\ \le & a_k, & k \in s_{g,\mathrm{INFL}} \end{array}. \tag{3.6}$$

The ordinary units and influential units constitute the units whose weights might be changed. Therefore we call them the *modifiable* units and write

$$s_{g,\mathrm{MOD}} = s_{g,\mathrm{ORD}} \uplus s_{g,\mathrm{INFL}} = s_g - s_{g,\mathrm{EXCL}}\,. \tag{3.7}$$

The identification of excluded units is done prior to any detection of influential units. Therefore the sets $s_{g,\mathrm{EXCL}}$ and $s_{g,\mathrm{MOD}}$ are fixed when the step of detecting influential behavior begins.

The user may also have a pre-specified list of excluded units $s_{g,\mathrm{EXCL,INIT}}$ and a pre-specified list of influential units $s_{g,\mathrm{INFL,INIT}}$. Any unit in $s_{g,\mathrm{EXCL,INIT}}$ is automatically put into $s_{g,\mathrm{EXCL}}$ and any unit in $s_{g,\mathrm{INFL,INIT}}$ is automatically put into $s_{g,\mathrm{INFL}}$.

Let n_g, $n_{g,\mathrm{EXCL}}$, $n_{g,\mathrm{ORD}}$, $n_{g,\mathrm{INFL}}$ and $n_{g,\mathrm{MOD}}$ be the numbers of units in s_g, $s_{g,\mathrm{EXCL}}$, $s_{g,\mathrm{ORD}}$, $s_{g,\mathrm{INFL}}$ and $s_{g,\mathrm{MOD}}$ respectively.

Processing influential units when the data group s_g is small could be risky. Suppose for example a data group s_g with $n_g = 4$. Then intuitively all four units have a big impact on the estimates $\hat{T}_{gj}$, so it does not make much sense to try flag some of them as influential. Therefore there exists some lower bound for the size of the data group, n_{MIN}. Any group with less than n_{MIN} units will not be processed. This means that the output weights of its units will be all set equal to the input weights. The value n_{MIN} is one of the parameters the user can control. However the value must be at least 5.

Actually, n_{MIN} is also the minimum size for $s_{g,\text{MOD}}$, the list of non-excluded units. Therefore the following holds:

- If $n_g < n_{\text{MIN}}$:
  - No identification of excluded units is done.
  - No detection of influential units is done.
  - The output weights are set equal to the input weights.
- If $n_{g,\text{MOD}} < n_{\text{MIN}} \leq n_g$:
  - Identification of excluded units is done.
  - No detection of influential units is done.
  - The output weights are set equal to the input weights.
- If $n_{g,\text{MOD}} \geq n_{\text{MIN}}$:
  - Identification of excluded units is done.
  - Detection of influential units is done.
  - If there are influential units, the full Module's algorithm runs.

4 Influential behavior

4.1 Local influential behavior

For a given set of weights $\{a_k\}$ and a variable of interest y_j, a unit $k \in s_{g,\text{MOD}}$ is to said to have a *local influential behavior* on the estimate $\hat{T}_{agj}$ if it has an excessive effect on its value.

The Influential Data Module offers four methods to quantify this effect.

Detection method	**A unit $k \in s_g$ is influential with respect to variable y_j and weights $\{a_k\}$ if**
AY	$z_{ajk} \equiv \text{sgn}(\hat{T}_{agj})\, a_k y_{jk} \geq \delta'_{gj}$ (4.1)
A|Y|	$z_{ajk} \equiv a_k \lvert y_{jk} \rvert \geq \delta'_{gj}$ (4.2)
TDIFF	$z_{ajk} \equiv \text{sgn}(\hat{T}_{agj})\,(\hat{T}_{agj} - \hat{T}_{agjk}) \geq \delta'_{gj}$ (4.3)
|TDIFF|	$z_{ajk} \equiv \lvert \hat{T}_{agj} - \hat{T}_{agjk} \rvert \geq \delta'_{gj}$ (4.4)

where δ'_{gj} is some upper bound properly chosen,

$$\operatorname{sgn}(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 1 \end{cases}$$

and $\hat{T}_{gajk}$ is the estimate we would get if unit k was removed from $s_{g,\mathrm{MOD}}$ and the remaining units were proportionally reweighted:

$$\hat{T}_{agjk} = \sum_{k \in s_{g,\mathrm{EXCL}}} a_k y_{jk} + \frac{\hat{N}_{a,g,\mathrm{MOD}}}{\hat{N}_{a,g,\mathrm{MOD}} - a_k} \sum_{l \in s_{g,\mathrm{MOD}-\{k\}}} a_l y_l, \tag{4.5}$$

where

$$\hat{N}_{ag,\mathrm{MOD}} = \sum_{k \in s_{g,\mathrm{MOD}}} a_k \,. \tag{4.6}$$

For each analysis variable y_j, the user must choose which detection method to use. Not all analysis variables need to use the same detection method, but it is not allowed to use more than one detection method per variable.

4.2 Global influential behavior

The definition of influential behavior of the previous section is relative to one analysis variable at a time (hence the adjective *local*). As the Module can process $J \geq 1$ analysis variables simultaneously, we need to have a definition of influential behavior irrelative to a specific analysis variable; otherwise the weight transformation would change with each analysis variable. The following is used:

Unit $k \in s_{g,\mathrm{MOD}}$ is said to have a *global influential behavior* if it has a local influential behavior for at least J_{MIN} analysis variables, where $1 \leq J_{\mathrm{MIN}} \leq J$ is some user-specified constant.

Note that this global influential behavior is still relative to the set of weights $\{a_k\}$ used. This is important to remember for the rest of this document.

4.3 Some comments on the detection methods

All the detection methods of section 4.1 generate some derived z variable and the condition of local influential behavior is based on $z_{ajk} \geq \delta'_{gj}$.

4.3.1 Detection method: AY

The AY detection method uses $z_{ajk} = \operatorname{sgn}(\hat{T}_{agj})\, a_k y_{jk}$. The quantity $a_k y_{jk}$ is the contribution of unit k to the estimate $\hat{T}_{gj}$. When this value is large (say 20% of $\hat{T}_{agj}$), it is considered influential. The use of $\operatorname{sgn}(\hat{T}_{agj})$ is to take in account the dominant sign of values of y_j. For example, if $\hat{T}_{gj} < 0$, the condition becomes

$$a_k y_{jk} \leq -\delta'_{gj}\,.$$

It is thus interpreted as a unit contributing a large value in the same direction as the estimate (large negative contribution if the estimate is negative, large positive contribution otherwise).

4.3.2 Detection method: A|Y|

The A|Y| detection method uses $z_{ajk} = a_k |y_k|$. It is a bilateral variant of the AY method: here values of y_j largely negative or largely positive are equally important.

4.3.3 Detection method: TDIFF

The TDIFF detection method uses $z_{ajk} = \operatorname{sgn}(\hat{T}_{agj})\,(\hat{T}_{agj} - \hat{T}_{agjk})$. TDIFF stands for *totals' difference*. It is based on the difference between the full estimate $\hat{T}_{agj}$ and the estimate $\hat{T}_{agjk}$ we would get if unit k was removed from s_g and the remaining units were reweighted to compensate.

Suppose for the sake of illustration that $\hat{T}_{agj} > 0$. Then, when $\operatorname{sgn}(\hat{T}_{agj})\,(\hat{T}_{agj} - \hat{T}_{agjk})$ is large, it means that removing the unit k decreases a lot the estimate, or inversely having the unit k in the sample increases a lot the estimate.

The same rationale stands when $\hat{T}_{agj} < 0$. Then $\operatorname{sgn}(\hat{T}_{agj})\,(\hat{T}_{agj} - \hat{T}_{agjk})$ being large means that $\hat{T}_{agj} - \hat{T}_{agjk}$ is small, which in return means that having the unit k in the sample decreases a lot the estimate.

The Method TDIFF relates to the usual definition of influential units in the context of regression: in that context, a unit is influential if it modifies greatly the beta coefficients of the regression. Here the betas are replaced by the totals $\hat{T}_{agj}$.

It is worthwhile that the Method TDIFF detects only units that increase the value of the estimate in the direction of the estimate, that is, it is a large positive value if the estimate is positive, or it is a large negative value if the estimate is negative. To detect units in both directions whatever the sign of the estimate, we will instead use the Method |TDIFF| in the next section.

4.3.4 Detection method: |TDIFF|

The TDIFF detection method uses $z_{ajk} = \left|\hat{T}_{agj} - \hat{T}_{agjk}\right|$. The |TDIFF| method is the bilateral version of the TDIFF method. Here, if $\left|\hat{T}_{agj} - \hat{T}_{agjk}\right|$ has a large value, it could either be because the unit dramatically increases the estimate or dramatically decreases the estimate, and this whatever the sign of the estimate.

4.4 Specifying the bound δ'_{gj}

We haven't so far said a word about how to specify the bounds δ'_{gj}.

Generally, the user specifies an alternative bound δ_{gj} and indicates the type of link between δ_{gj} and δ'_{gj}.

Delta type	Relation
ABS	$\delta'_{gk} = \delta_{gj}$
REL	$\delta'_{gk} = \delta_{gj}\left\|\hat{T}_{gj}\right\|$
RELN	$\delta'_{gk} = \frac{\delta_{gj}\left\|\hat{T}_{gj}\right\|}{n_g}$
QUARTILE	$\delta'_{gj} = (Q3)_{z_{aj}} + \delta_{gj}(IQR)_{z_{aj}}$

The ABS type, for *absolute*, indicates that the δ_{gj} is directly the δ'_{gj}. One must take additional care when using the ABS type, as it is the user's responsability to adapt the value of each δ'_{gj} to take into account the scale of variable y_j and the number of units n_g in the data group s_g.

The REL type, for *relative*, takes care of the scale of variable y_j. However it does not take care of the size n_g of the data group.

The RELN type takes care of both the scale of y_j and of the size of the data group.

Finally the QUARTILE type is likely the most adaptive way to build the δ'_{gj}'s. It is based on Tukey's quartile method, which is used for instance in the box-plot.

For the variable z_{aj}, one calculates the third quartile, $(Q3)_{z_{aj}}$, and the interquartile distance, $(IQR)_{z_{aj}}$, within the data group s_g. Using 1.5 for δ_{gj} corresponds to the near fences in the box-plot and using 3 corresponds to the far fences. Therefore an intuitive value for δ_{gj} for the QUARTILE type could be 3.

5 Step 1: Initialization of the data group

At the very beginning of its execution, the Module initializes the creation of the three subsets $s_{g,\mathrm{EXCL}}$, $s_{g,\mathrm{ORD}}$ and $s_{g,\mathrm{INFL}}$ from s_g.

If the user has an initial list of excluded units $s_{g,\mathrm{EXCL,INIT}}$, then $s_{g,\mathrm{EXCL}} \coloneqq s_{g,\mathrm{EXCL,INIT}}$. Otherwise $s_{g,\mathrm{EXCL}} \coloneqq \emptyset$.

If the user has an initial list of influential units $s_{g,\mathrm{INFL,INIT}}$, then $s_{g,\mathrm{INFL}} \coloneqq s_{g,\mathrm{INFL,INIT}}$. Otherwise, $s_{g,\mathrm{INFL}} \coloneqq \emptyset$.

Note that by design, these two initial lists cannot intersect. Indeed, the initial lists are derived from a special flag variable on the sample file, called an input unit type variable. A value equal to “x” means a pre-specified excluded unit. A value equal to “i” means a pre-specified influential unit. A value equal to “o” means a pre-specified ordinary unit.

Let $s_{g,ORD} \coloneqq s_g - (s_{g,\mathrm{EXCL}} \cup s_{g,\mathrm{INFL}})$.

6 Step 2: Identifying excluded units

Excluded units in the Influential Data Module are units that are prohibited from a weight change. So they cannot be flagged as influential, and they cannot also be classified as ordinary units, because ordinary units could have their weight increased in order to compensate for influential units whose weights would be decreased.

A typical example of excluded units could be units with an input weight equal to 1 (*take-all*).

Excluded units are not mandatory to exist for the Influential Data Module. In fact, unless some specific parameters of the macro are specified, the Module will assume that all units' weights can be modified. It is thus the user's responsibility to properly identify excluded units, if (s)he believes that some units should never be changed.

There are five ways to identify excluded units in the Module.

- Pre-specified list
- Lower bound for the input weight
- Boolean condition
- Proportion of smallest values of the *z* variables
- Lower bound for the *z* variables

6.1 Excluding a pre-specified list

The user has himself identified a list of units to be excluded. He let the Module be aware of this list by creating an input unit type variable whose value “x” means a pre-specified excluded unit.

Let’s denote this list $s_{\mathrm{EXCL,INIT}}$ and let $s_{g,\mathrm{EXCL,INIT}} = s_g \cap s_{\mathrm{EXCL,INIT}}$ be its part in s_g.

6.2 Excluding with a lower bound for the input weight

The user specifies some lower bound a_{EXCL} and units k with $a_k \leq a_{\text{EXCL}}$ are excluded. Let $s_{g,\text{EXCL,LW}}$ denote these units from s_g; LW stands for "lower bound".

For instance, one could specify $a_{\text{EXCL}} = 1$, in order to exclude take-alls.

6.3 Excluding with a Boolean expression

The user specifies some Boolean expression and any unit meeting that condition is excluded.

For example, if this Boolean expression is

FLAG1=1 or FLAG2 in (‘A’, ‘B’)

then any unit having FLAG1=1 or FLAG2=’A’ or FLAG3=’B’ is excluded. Here FLAG1 and FLAG2 must be existing variables on the sample file.

Let’s $s_{g,\text{EXCL,EXPR}}$ be the part of s_g that is excluded by the Boolean expression criterion.

6.4 Excluding some proportion of smallest values of the z variables

We consider the following derived variables z_{ajk}:

Detection method	Z variable	
AY	$z_{ajk} \equiv \text{sgn}(\hat{T}_{agj})\, a_k y_{jk}$	(6.1)
A|Y|	$z_{ajk} \equiv a_k \lvert y_{jk} \rvert$	(6.2)
TDIFF	$z_{ajk} \equiv \text{sgn}(\hat{T}_{agj})\, (\hat{T}_{agj} - \hat{T}_{agjk0})$	(6.3)
|TDIFF|	$z_{ajk} \equiv \lvert \hat{T}_{agj} - \hat{T}_{agjk0} \rvert$	(6.4)

where

$$\hat{T}_{ajk0} = \frac{\hat{N}_{ag}}{\hat{N}_{ag} - a_k} \sum_{l \in s_{g-\{k\}}} a_l y_l \tag{6.5}$$

and

$$\widehat{N}_{ag} = \sum_{k \in s_g} a_k \, . \tag{6.6}$$

Note that the *z* variable for the method TDIFF and |TDIFF| is slightly different from the one in section 4.1, as there is not yet a list of excluded units here. (Remember that the identification of excluded units is done before the detection of influential units).

Let p_{gj} some number in the interval [0,1], $1 \le g \le G$, $1 \le j \le J$. Not all J variables need to be active for this exclusion criterion. When a variable y_j is not active, it means that we don't have a p_{gj} for any g.

Let A be the list of active indices j.

For an active index $j \in A$, for each g:

The values z_{jk} are calculated. They are sorted in ascending number and unduplicated:

$$z_{\bullet gj1} < z_{\bullet gj2} < \cdots z_{\bullet gjL} \, .$$

Let B_{gjl}be the aggregate having the value $z_{\bullet gjl}$: $B_{gjl} = \{k \in s_g : z_{gjk} = z_{\bullet gjl}\}$. Let m_{gjl} be the number of units in the aggregate B_{gjl}.

The cumulative frequence is calculated:

$$M_{gjl} = \sum_{l' \le l} m_{gjl} \, , \qquad 1 \le l \le L, \qquad 1 \le g \le G.$$

M_{gjl} is thus the number of units k of s_g whose value z_{jk} is less than or equal to $z_{\bullet gjl}$.

Let

$$s_{g,\mathrm{EXCL,PROP},j} = \bigcup_{l : M_{gjl} \le p_{gj} n_g} B_{gjl} \, .$$

Therefore $s_{g,\mathrm{EXCL,PROP},j}$ is the list of the units whose value of z_{jk} is less than or equal to the $100p_{gj}$-th quantile. The unduplication above is to make sure that two units with the same value for the variable z do not get separated into the set of excluded and the set of non-excluded.

Once the files $s_{g,\mathrm{EXCL,PROP},j}$ have been calculated for each active index $j \in A$, the actual list of excluded units for the current exclusion criterion is

$$s_{g,\mathrm{EXCL,PROP}} = \bigcap_{j \in A} s_{g,\mathrm{EXCL,PROP},j} \, .$$

6.5 Excluding with a lower bound for the z variables

Let's reuse the z variables introduced in section 6.4.

Let δ'_{gj} some lower bound, $1 \leq g \leq G$, $1 \leq j \leq J$. Not all J variables need to be active for this exclusion criterion. When a variable y_j is not active, it means that we don't have a δ_{gj} for any g.

Let A be the list of active indices j. For active variables for this exclusion criterion, let

$$s_{g,\mathrm{EXCL,DELTA},j} = \{k \in s_g : z_{jk} \leq \delta'_{gj}\}, \qquad 1 \leq g \leq G, j \in A\,.$$

The list of units excluded with the current criterion is

$$s_{g,\mathrm{EXCL,DELTA}} = \bigcap_{j \in A} s_{g,\mathrm{EXCL,DELTA},j}\,.$$

The bound δ'_{gj} can be directly or indirectly specified. In the general case, the user specifies a bound δ_{gj} and indicates the link between δ_{gj} and δ'_{gj} as follows:

Delta type	**Relation**
ABS	$\delta'_{gk} = \delta_{gj}$
REL	$\delta'_{gk} = \delta_{gj}\lvert\hat{T}_{gj}\rvert$
RELN	$\delta'_{gk} = \frac{\delta_{gj}\lvert\hat{T}_{gj}\rvert}{n_g}$

6.6 Combining the various exclusion criteria

The final list of excluded data is the union of the lists obtained for each criterion of the five sections above:

$$s_{g,\mathrm{EXCL}} = s_{g,\mathrm{EXCL,INIT}} \cup s_{g,\mathrm{EXCL,LW}} \cup s_{g,\mathrm{EXCL,EXPR}} \cup s_{g,\mathrm{EXCL,PROP}} \cup s_{g,\mathrm{EXCL,DELTA}}\,.$$

Note that not all five criteria need to be specified. When a criterion is not specified, the corresponding set is empty. When none of the five criteria is used, then $s_{g,\mathrm{EXCL}}$ is the empty set, that is, there are no excluded units.

Once $s_{g,\mathrm{EXCL}}$ is known, we define

$$s_{g,\mathrm{MOD}} = s_g - s_{g,\mathrm{EXCL}}\,.$$

7 Step 3: Detection of influential behavior, using the input weights

After the optional identification of excluded units, comes the first detection of influential units.

If an initial list of influential units $s_{g,\mathrm{INFL,INIT}}$ was provided, the detection of influential units is optional. Running it can only increase the pool of influential units $s_{g,\mathrm{INFL}}$.

If however no $s_{g,\mathrm{INFL,INIT}}$ was specified, the detection of influential units is mandatory.

The first detection is done with the input weights $\{a_k\}$. The z_{ajk} used for the detection are those in section 4.1, using the sets $s_{g,\mathrm{EXCL}}$, $s_{g,\mathrm{ORD}}$ and $s_{g,\mathrm{INFL}}$ as they are at this point.

If a unit $k \in s_{g,\mathrm{ORD}}$ displays a global influential behavior, it is transferred from $s_{g,\mathrm{ORD}}$ to $s_{g,\mathrm{INFL}}$.

No unit from $s_{g,INFL}$ is transferred to $s_{g,\mathrm{ORD}}$, even if it does not exhibit a global influential behavior. This means that if a unit is in an initial list of influential units (section 5), it will remain flagged as influent, whatever the detection step says about it.

8 Step 4: Iterative process to find the optimal weight transformation

This step is run only if $s_{g,\mathrm{INFL}}$ is not empty. If it is empty, then the Module considers that the weight transformation is the identity function, that is, the output weights will be equal to the input weights for all units in the data group s_g.

The Module goes into an iterative process where a different weight transformation is attempted at each iteration. The resulting test weights, $\{w_k\}$, are evaluated with regard to three criteria: (a) they do not generate influential behavior; (b) their corresponding estimates $\hat{T}_{wgj}$ are not too much biased; and (c) the mean squared error (MSE) of the estimates $\hat{T}_{wgj}$ is smaller to some degree than the MSE of the initial estimates $\hat{T}_{agj}$.

We will first describe the type of weight transformation. Then we will explain the three criteria. We will explain some quantities (relative bias, relative mean error, etc.) that are derived at each iteration. Finally we will see how the iterative process works.

8.1 Weight transformation

The new weights w_k are derived from the input weights a_k through the following weight transformation:

$$w_k = \begin{cases} a_k, & k \in s_{g,\mathrm{EXCL}} \\ c_g a_k, & k \in s_{g,\mathrm{ORD}} \\ a_{k\,g}^{\;e_g}, & k \in s_{g,\mathrm{INFL}} \end{cases} . \tag{8.1}$$

The only true parameter of this transformation is the exponent e_g to be applied to the influential units. This exponent will be in the interval]0,1[in order to reduce the weights of units in $s_{g,\mathrm{INFL}}$ (remember that the assumption is that all input weights $a_k \geq 1$). Once this exponent is known, the multiplier c_g to be applied to ordinary units is chosen as to preserve the weight at the data group level:

$$\sum_{k \in s_g} w_k = \sum_{k \in s_g} a_k . \tag{8.2}$$

This entails

$$c_g = 1 + \frac{\sum_{k \in s_{g,\mathrm{INFL}}} \left(a_k - a_k^{e_g}\right)}{\sum_{k \in s_{g,\mathrm{ORD}}} a_k} . \tag{8.3}$$

Remark:

Applying an exponent less than one to a weight initially greater than one will bring this weight closer to one, but not necessarily equal to it (unless the exponent is itself zero). So the weight transformation of the current Module is a compromise between resetting the weight of influential units to one, seen as too drastic, and leaving the weight unchanged. Moreover, if two influentail units k and l are such that $1 < a_k < a_l$, then $a_k^{e_g} < a_l^{e_g}$, that is, the weight ranks do not change amon g influential units after the weight transformation. Similarly, it is easy to show that the multiplicative coefficient c_g is positive and thus the weigth ranks do not change among ordinary units after the weight transformation.

8.2 Criteria on the test weights

Each time a new set of test weights $\{w_k\}$ is created, the Module evaluates them against some active criteria. There are three criteria. The user can specify which ones are to be used, except for the following restriction: if the user did not provide a pre-specified list of influential units $s_{g,\mathrm{INFL,INIT}}$, the influential criterion must be active. If such an initial list exists, then the user may skip the influential criterion.

8.2.1 The influential criterion

The influential criterion consists of detecting influential behavior, as in section 7, except this time the test weights $\{w_k\}$ are used instead of the input weights $\{a_k\}$.

When this criterion is evaluated, the following subsets are generated:

$s_{g,\text{ii}}$	Units of $s_{g,\text{INFL}}$ that have a global influential behavior with the test weights
$s_{g,\text{io}}$	Units of $s_{g,\text{INFL}}$ that doesn't have a global influential behavior with the test weights
$s_{g,\text{oi}}$	Units of $s_{g,\text{ORD}}$ that have a global influential behavior with the test weights
$s_{g,\text{oo}}$	Units of $s_{g,\text{ORD}}$ that doesn't have a global influential behavior with the test weights

Of course

$$s_{g,\text{INFL}} = s_{g,\text{io}} \cup s_{g,\text{ii}},$$

$$s_{g,\text{ORD}} = s_{g,\text{oo}} \cup s_{g,\text{oi}}.$$

The influential criterion is satisfied if $s_{g,\text{ii}}$ and $s_{g,\text{oi}}$ are empty, that is, the test weights do not generate any global influential behavior. If this is not the case, we either have $s_{g,\text{ii}} \neq \emptyset$ or $s_{g,oi} \neq \emptyset$ or both. The event $s_{g,\text{ii}} \neq \emptyset$ typically means that the weight transformation was too *mild* and was not able the remove the influential behavior observed with the input weights. On the other hand, if the weight transformation is too strong, we may end up by reducing a lot the weights of units in $s_{g,\text{INFL}}$, but at the detriment of increasing the weights of units in $s_{g,ORD}$ and thus generating influential behavior among these units, when there was none to start with.

Sometimes it is impossible to find a weight transformation that will remove the influential behavior from units in $s_{g,\text{INFL}}$ without generating some for units in $s_{g,\text{ORD}}$. If this is such the case, it means we have *collateral* influential units, that is, units that were not influential per so with the input weights, but were very close to be, and any weight transformation to correct the influential units will transform these units from $s_{g,\text{ORD}}$ into influential ones. The Module then transfers these units from $s_{g,\text{ORD}}$ to $s_{g,\text{INFL}}$, and the iterative process restarts.

It is important to understand that once a unit is placed in $s_{g,\text{INFL}}$ it cannot be moved from it. Being in this set means that at some point (either at the very beginning, with the input weights a_k, or at some iteration, with the test weights) these units exhibited an influential behavior and the Module has no choice as to reduce their weights. Units can thus transit from $s_{g,\text{ORD}}$ to $s_{g,\text{INFL}}$ but never in the other direction.

Example showing how the four subsets may behave in terms of the exponent tried

Exponent	$\text{card}(s_{g,\text{io}})$	$\text{card}(s_{g,\text{ii}})$	$\text{card}(s_{g,\text{oo}})$	$\text{card}(s_{g,\text{oi}})$
1.00	0	5	100	0
0.95	1	4	100	0
0.90	2	3	100	0
0.85	3	2	99	1
0.80	4	1	99	1
0.75	5	0	98	2
0.70	5	0	97	3
0.65	5	0	95	5

0.60	5	0	92	8

The number of units in $s_{g,\text{INFL}}$ is 5 and the number of units in $s_{g,\text{ORD}}$ is 100. When the exponent is 1, it means that no change is done to the weights and thus all units in in $s_{g,\text{INFL}}$ still have an influential behavior, while all units in $s_{g,\text{ORD}}$ still have a "normal" behavior. When exponent 0.95 is tried, there is one unit from $s_{g,INFL}$ that does not have an influential behavior; the four other ones still behave badly. This is not surprising, as the exponent is still close to 1 and thus does not induce too much change in the weights.
However, as the exponent decreases, the weights of units in $s_{g,\text{INFL}}$ decrease and they stop behaving in an influential way. However, the weights of units in $s_{g,\text{ORD}}$ increase and they start having an influential behavior.

In this example, there is no exponent such that $\text{card}(s_{g,\text{ii}}) = \text{card}(s_{g,\text{oi}}) = 0$. That means that new units will need to be added to the set $s_{g,\text{INFL}}$.The largest exponent for which $\text{card}(s_{g,\text{ii}}) = 0$ is 0.75. At this exponent, $\text{card}(s_{g,\text{oi}}) = 2$. So there are two units we need to be transferred from $s_{g,\text{ORD}}$ to $s_{g,\text{INFL}}$. Once the lists $s_{g,\text{ORD}}$ and $s_{g,\text{INFL}}$ have been changed, the Module will need to restart the evaluation of the various exponents.

8.2.2 The bias criterion

The current criterion tries to minimize the bias introduced in the estimators when the input weights a_k are replaced with the new weights w_k.

In order to estimate that bias, we make the following assumption and approximation: (a) the estimates $\hat{T}_{agj}$ obtained with the input weights, especially if those are the design weights or close to them, are unbiased or almost unbiased, so we assume the bias of $\hat{T}_{agj}$ is zero; (b) the estimates $\hat{T}_{wgj}$ are possibly biased and we estimate their bias $B(\hat{T}_{wgj}) = E(\hat{T}_{wgj}) - T_{gj}$ with

$$\hat{B}(\hat{T}_{wgj}) = \hat{T}_{wgj} - \hat{T}_{agj}, \qquad 1 \le g \le G, 1 \le j \le J\,.$$

The bias criterion consists of setting a upper bound δ'_{gj} on the bias:

$$\left|\hat{B}(\hat{T}_{wgj})\right| \le \delta'_{gj}, \qquad 1 \le g \le G, 1 \le j \le J\,.$$

Not all variables y_j need to be active for the bias criterion. Let A be the set of active indices j: $A \subset \{1,2,\dots,J\}$. When a variable y_j is active, it means that δ'_{gj} is specified for each data group s_g.

The bias criterion is satisfied when the relation above holds with the test weights w_k, for all active j.

It is possible not to evaluate at all the bias criterion. This means that the user is not interested in this criterion.

The upper bound δ'_{gj} can be directly or indirectly specified by the user. The user specifies a bound δ_{gj} and indicates the link between δ_{gj} and δ'_{gj}:

Delta type	Relation
ABS	$\delta'_{gy} = \delta_{gy}$

REL	$\delta'_{gy} = \delta_{gy} \lvert \hat{T}_{agy} \rvert$

8.2.3 The MSE criterion

While the new estimators $\hat{T}_{wgj}$ are seen more biased than the initial estimators $\hat{T}_{agj}$, they should make amends by reducing the mean squared error (MSE). The MSE criterion is thus a way to ensure that this improvement is effective.

Let A be the set of indices of variables y_j to which one wants to apply the MSE criterion. For each such so-called active variable y_j,let δ_{gj} be some number in the interval [0,1], $1 \leq g \leq G$.

The MSE criterion is satisfied if

$$\widehat{MSE}(\hat{T}_{wgj}) \leq \delta_{gj} \widehat{MSE}(\hat{T}_{agj}), \qquad 1 \leq g \leq G, j \in A$$

where $\widehat{MSE}$ denotes an estimate of the mean squared error.

In order to estimate the MSE, two methods are available:

- The naïve method, where Taylorization is done.
- The bootstrap method.

In both methods, one can specify strata, which may differ from the data groups U_g. Therefore, in general the MSE criterion cannot be processed separately for each data group s_g; instead it invoves the full sample.

In the case of the bootstrap method, the algorithm is as follows:

1. B_{MSE} Bootstrap replicates are generated from the input weights a_k, using the Rao-Wu approach. The number B_{MSE} is specified by the user. These bootstrap replicates are not the same as the input bootstrap replicates $a_{(b)k}$. Let $a_{\text{MSE},(b)k}$ be the bootstrap weight of unit k for the replicate number$b$, $k \in s$, $1 \leq b \leq B_{\text{MSE}}$. Note that these bootstrap weights are not generated at each iteration. They are generated only once, before the iterative process.
2. The classification of influential, excluded and ordinary units is not redone for each bootstrap replicate. We accept the classification done with the input non-bootstrap weights a_k: $s_{g,\text{INFL}}$, $s_{g,\text{EXCL}}$ and $s_{g,\text{ORD}}$.
3. Let's define $w_{\text{MSE},(b)k}$ as the new weight resulting from applying the weight transformation to $a_{\text{MSE},(b)k}$. The main parameter of the weight transformation, the exponent e_g, $1 \leq g \leq G$, to be applied to the weight of influential units, was based on the input weights a_k and is not recalculated for each bootstrap replicate. However the coefficient c_g for ordinary units is recalculated as $c_{g(b)}$ for each replicate.
   More precisely:
   If $k \in s_{g,\text{EXCL}}$, then $w_{\text{MSE},(b)k} = a_{\text{MSE},(b)k}$.
   If $k \in s_{g,\text{INFL}}$, then $w_{\text{MSE},(b)k} = a_{\text{MSE},(b)k}^{e_g}$.

If $k \in s_{g,ORD}$, then $w_{MSE,(b)k} = c_{g(b)} a_{\text{MSE},(b)k}$
where

$$c_{g(b)} = 1 + \frac{\sum_{k \in s_{g,\text{INFL}}} \left(a_{\text{MSE},(b)k} - w_{\text{MSE},(b)k} \right)}{\sum_{k \in s_{g,\text{ORD}}} a_{\text{MSE},(b)k}}.$$

4. Let

$$\hat{T}_{agj(b)} = \sum_{k \in s_g} a_{\text{MSE},(b)k},$$

$$\hat{T}_{wgj(b)} = \sum_{k \in s_g} w_{\text{MSE},(b)k}.$$

5. The mean squared error of the estimator $\hat{T}_{agj}$ is calculated as

$$\widehat{MSE}\left(\hat{T}_{agj}\right) = \frac{1}{B-1} \sum_{b=1}^{B} \left(\hat{T}_{agj(b)} - \hat{T}_{agj(\bullet)} \right)^2$$

where

$$\hat{T}_{agj(\bullet)} = \frac{1}{B} \hat{T}_{agj(b)}.$$

The assumption here is that the bias of $\hat{T}_{agj}$ is zero, so that the mean squared error is equal to the variance.

6. The mean squared error of the estimator $\hat{T}_{wgj}$ is calculated as

$$\widehat{MSE}\left(\hat{T}_{agj}\right) = \left(\hat{T}_{wgj} - \hat{T}_{agj} \right)^2 + \frac{1}{B-1} \sum_{b=1}^{B} \left(\hat{T}_{wgj(b)} - \hat{T}_{wgj(\bullet)} \right)^2$$

where

$$\hat{T}_{wgj(\bullet)} = \frac{1}{B} \hat{T}_{wgj(b)}.$$

8.3 Quantities derived

The following quantities are calculated each iteration.

Name	Definition
A_{infl}	List of indices j of analysis variables active for the influential criterion. $A_{\text{infl}} \subset \{1, 2, \ldots, J\}$.
A_{bias}	List of indices j of analysis variables active for the bias criterion. This set can be empty if the user does not want to use the bias criterion. $A_{\text{bias}} \subset \{1, 2, \ldots, J\}$.
A_{MSE}	List of indices j of analysis variables active for the MSE criterion. This set can be empty if the user does not want to use the MSE criterion. $A_{\text{MSE}} \subset \{1, 2, \ldots, J\}$.
Absolute value of the relative bias of $\hat{T}_{wgj}$	$\left\lvert \widehat{RB}_{gj} \right\rvert = \frac{\left\lvert \hat{T}_{wgj} - \hat{T}_{agj} \right\rvert}{\left\lvert \hat{T}_{agj} \right\rvert}, \quad 1 \le j \le J$

Sum of the absolute values of the relative biases	$$\lvert\widehat{RB}_g\rvert = \sum_{j\in A_{\text{bias}}} \widehat{RB}_{wgj}$$
Number of variables for which the univariate bias criterion condition is met	$$C_{g,\text{bias}} = \sum_{j\in A_{\text{bias}}} I\left(\left\lvert\hat{B}(\hat{T}_{wgj})\right\rvert \le \delta'_{gj}\right)$$ where $$I(p) = \begin{cases}1, & p \text{ is true}\\ 0, & p \text{ is false}\end{cases}$$
Relative mean error of $\hat{T}_{agj}$	$$\widehat{RME}_{agj} = \frac{\sqrt{\widehat{MSE}_{agj}}}{\lvert\hat{T}_{agj}\rvert}, \qquad j \in A_{\text{MSE}}$$
Relative mean error of $\hat{T}_{wgj}$	$$\widehat{RME}_{wgj} = \frac{\sqrt{\widehat{MSE}_{wgj}}}{\lvert\hat{T}_{agj}\rvert}, \qquad j \in A_{\text{MSE}}$$ (Note that the denominator is $\lvert\hat{T}_{agj}\rvert$, in order to make comparable $\widehat{RME}_{agj}$ and $\widehat{RME}_{wgj}$).
Ratio of the MSEs for the estimates of T_{gj}	$$\widehat{RMSE}_{gj} = \frac{\widehat{MSE}_{wgj}}{\widehat{MSE}_{agj}}, \qquad j \in A_{\text{MSE}}$$
Number of variables for which the MSE criterion condition is met	$$C_{g,\text{MSE}} = \sum_{j\in A_{\text{MSE}}} I\left(\widehat{MSE}(\hat{T}_{wgj}) \le \delta_{gj}\widehat{MSE}(\hat{T}_{agj})\right)$$
Sum of the relative errors of $\hat{T}_{agj}$	$$\widehat{RME}_{ag} = \sum_{j\in A_{\text{MSE}}} \widehat{RME}_{agj}$$
Sum of the relative errors of $\hat{T}_{wgj}$	$$\widehat{RME}_{wg} = \sum_{j\in A_{\text{MSE}}} \widehat{RME}_{wgj}$$

The bias criterion is met if and only if $C_{g,\text{bias}} = \text{card}(A_{\text{bias}})$, where card is the set function returning the number of elements in the set. The MSE criterion is satisfied if and only if $C_{g,\text{MSE}} = card(A_{\text{MSE}})$,

8.4 Description of the iterations

The iterative nature of the algorithm depends on the way the exponents are selected. The user is given two choices:

- Either he provides a finite list of exponents
- Or he asks the Module to use a bisection algorithm to choose the exponents

8.4.1 Finite list of exponents

If the exponent selection is a list of user-defined exponents:

1. Try each exponent in the list. It means generating the test weights and evaluating if the active criteria are met. No change of classification of units is done at this stage.
2. If there are exponents such that (a) $s_{g,\text{ii}} = s_{g,\text{oi}} = \emptyset$, (b) the MSE criterion is passed and (c) the the bias criterion is passed, then:
   a. Sort these exponents by
      i. increasing relative mean error ($\widehat{RME}_{wg}$),
      ii. increasing absolute value of the relative bias ($|\widehat{RB}_g|$),
      iii. decreasing exponent.
   b. Pick the first exponent and let it be the final exponent for this data group. Exit.
3. If there are exponents such that (a) $s_{g,\text{ii}} = s_{g,\text{oi}} = \emptyset$ and (b) the MSE criterion is passed, then:
   a. Sort these exponents by
      i. decreasing $C_{g,\text{bias}}$,
      ii. increasing $|\widehat{RB}_g|$,
      iii. increasing $\widehat{RME}_{wg}$,
      iv. decreasing exponent.
   b. Pick the first exponent and let it be the final exponent for this data group. Exit.
4. If there are exponents such that (a) $s_{g,\text{ii}} = s_{g,\text{oi}} = \emptyset$ and (b) the bias criterion is passed, then:
   a. Sort these exponents by:
      i. decreasing $C_{g,\text{MSE}}$,
      ii. increasing $\widehat{RME}_{wg}$,
      iii. increasing $|\widehat{RB}_g|$,
      iv. decreasing exponent.
   b. Pick the first exponent and let it be the final exponent for this data group. Exit.
5. If there are exponents such that (a) $s_{g,\text{ii}} = s_{g,\text{oi}} = \emptyset$ and (b) the bias criterion is passed, then:
   a. Sort these exponents by:
      i. decreasing $C_{g,\text{MSE}}$,
      ii. decreasing $C_{g,\text{bias}}$,
      iii. increasing $\widehat{RME}_{wg}$,
      iv. increasing $|\widehat{RB}_g|$,
      v. decreasing exponent.
   b. Pick the first exponent and let it be the final exponent for this data group. Exit.
6. If there are exponents such that (a) $s_{g,\text{ii}} = \emptyset$ and (b) $s_{g,\text{oi}} \neq \emptyset$, then:
   a. Sort these exponents by:
      i. decreasing exponent.
   b. For the first exponent of the sort, transfer then units of $s_{g,\text{oi}}$ from $s_{g,\text{ORD}}$ to $s_{g,\text{INFL}}$.
   c. Go back to step 1.
7. (This step is executed only if all exponents were bad).
   a. Sort all exponents by:
      i. increasing $\text{card}(s_{g,\text{ii}})$,
      ii. decreasing $C_{g,\text{MSE}}$,

iii. decreasing $C_{g,\text{bias}}$,
iv. increasing $\widehat{RME}_{wg}$,
v. increasing $|\widehat{RB}_g|$,
vi. increasing $\text{card}(s_{g,\text{io}})$,
vii. decreasing exponent.

b. Pick the first exponent and let it be the final exponent for this data group. Exit.

8.4.2 Bisection algorithm for the exponents

The bisection algorithm related to an exponent can be described as follows: suppose we look for the largest exponent that fulfills some criterion C.

1. Let the lower bound $e_{\text{MIN}} \coloneqq 0$ and the upper bound $e_{\text{MAX}} \coloneqq 1$ for the possible exponents.
2. Let $e_{\text{MED}} \coloneqq (e_{\text{MIN}} + e_{\text{MAX}})/2$.
3. If $|e_{\text{MAX}} - e_{\text{MIN}}| < \varepsilon$ where ε is some predefined small number greater than zero:
   a. Let $e \coloneqq e_{\text{MED}}$. e is the exponent found.
   b. Exit the bisection algorithm.
4. Generate the test weights with this exponent and evaluate if the criterion C is satisfied.
5. If the criterion C is satisfied:
   a. Let $e_{\text{MIN}} \coloneqq e_{\text{MED}}$.
6. If the criterion C is not satisfied:
   a. Let $e_{\text{MAX}} \coloneqq e_{\text{MED}}$.
7. Go back to step 2.

If we look to the smallest exponents that fulfills the criterion C, we simply replace steps 5 and 6 above by

5. If the criterion C is satisfied:
   a. Let $e_{\text{MAX}} \coloneqq e_{\text{MED}}$.
6. If the criterion C is not satisfied:
   a. Let $e_{\text{MIN}} \coloneqq e_{\text{MED}}$.

Now the full algorithm is:

1. If the influential criterion is active:
   a. Find the largest exponent for which $s_{g,\text{ii}} = \emptyset$. Let $e_{\text{infl,MAX}}$ be this exponent.
   b. If for the exponent $e_{\text{infl,MAX}}$ we have $s_{g,\text{oi}} \neq \emptyset$:
      i. Transfer units of $s_{g,\text{oi}}$ from $s_{g,\text{ORD}}$ to$s_{g,\text{INFL}}$.
      ii. Go back to tep 1.
   c. Find the smallest exponent for which $s_{g,\text{ii}} = s_{g,\text{oi}} = \emptyset$. Let $e_{\text{infl,MIN}}$ be this exponent.
2. If the influential criterion is not active:
   a. Let $e_{\text{infl,MAX}} \coloneqq 1$.
   b. Let $e_{\text{infl,MIN}} \coloneqq 0$.
3. If the bias criterion is active:

a. Find the lowest exponent for which the bias criterion is satisfied. Let $e_{\text{bias,MIN}}$ be this exponent.

4. If the bias criterion is not active:
   a. Let $e_{\text{bias,MIN}} \coloneqq 0$.
5. If the MSE criterion is active:
   a. Find the largest exponent for which the MSE criterion is satisfied. Let $e_{\text{mse,MAX}}$ be this exponent.
   b. Find the smallest exponent for which the MSE criterion is satisfied. Let $e_{\text{mse,MIN}}$ be this exponent.
6. If the MSE criterion is not active:
   a. Let $e_{\text{mse,MAX}} \coloneqq 1$.
   b. Let $e_{\text{mse,MIN}} \coloneqq 0$.
7. Let
   a. $e_{\text{MIN}} \coloneqq \max(e_{\text{infl,MIN}}, e_{\text{bias,MIN}}, e_{\text{mse,MIN}})$,
   b. $e_{\text{MAX}} \coloneqq \min(e_{\text{infl,MAX}}, e_{\text{mse,MAX}})$.
8. If $\max(e_{\text{infl,MIN}}, e_{\text{bias,MIN}}, e_{\text{mse,MIN}}) \le \min(e_{\text{infl,MAX}}, e_{\text{mse,MAX}})$:
   a. Choose $\min(e_{\text{infl,MAX}}, e_{\text{mse,MAX}})$ as the final exponent for the data group. Exit.
9. If $\max(e_{\text{infl,MIN}}, e_{\text{mse,MIN}}) \le \min(e_{\text{infl,MAX}}, e_{\text{mse,MAX}})$:
   a. Choose $\min(e_{\text{infl,MAX}}, e_{\text{mse,MAX}})$ as the final exponent for the data group. Exit.
10. If $\max(e_{\text{infl,MIN}}, e_{\text{bias,MIN}}) \le e_{\text{infl,MAX}}$:
    a. Choose $e_{\text{infl,MAX}}$ as the final exponent for the data group. Exit.

9 Step 5: Create the output weights

When the step 4 (section 8) is done, the subsets $s_{g,\text{EXCL}}$, $s_{g,\text{ORD}}$ and $s_{g,\text{INFL}}$ are frozen and the final exponent e_g is found. Note that this exponent can be 1, as $s_{g,\text{INFL}}$ can be empty.

The output weights, w_k, are simply

$$w_k = \begin{cases} a_k, & k \in s_{g,\text{EXCL}} \\ c_g a_k, & k \in s_{g,\text{ORD}} \\ a_k^{e_g}, & k \in s_{g,\text{INFL}} \end{cases}$$

where a_k is the input weight and

$$c_g = 1 + \frac{\sum_{k \in s_{g,\text{INFL}}} \left(a_k - a_k^{e_g}\right)}{\sum_{k \in s_{g,\text{ORD}}} a_k}.$$

If in addition there were input bootstrap weights $a_{(b)k}$, the output bootstrap weights $w_{(b)k}$ are simply

$$w_{(b)k} = \left(\begin{matrix} a_{(b)k}, & k \in s_{g,\text{EXCL}} \\ c_{(b)g} a_{(b)k}, & k \in s_{g,\text{ORD}} \\ a_k^{e_g}, & k \in s_{g,\text{INFL}} \end{matrix} \right.$$

where

$$c_{(b)g} = 1 + \frac{\sum_{k \in s_{g,\mathrm{INFL}}} \left(a_{(b)k} - a_{(b)k}^{e_g} \right)}{\sum_{k \in s_{g,\mathrm{ORD}}} a_{(b)k}}.$$